

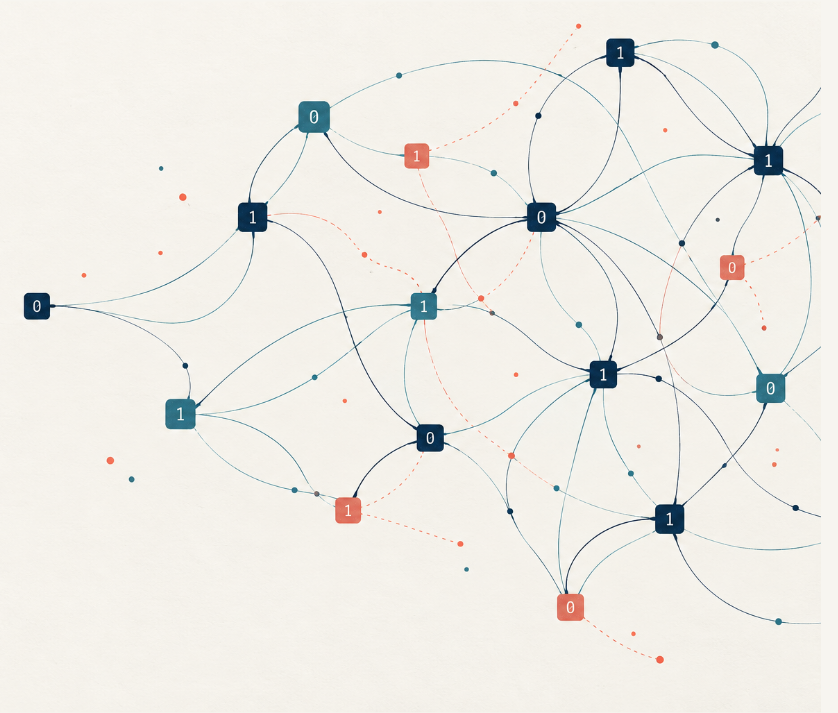
RSA, Kyber and the next public-key scheme

A case study for an ABW research programme

Practical appeal, security evidence and design tradeoffs

Cryptography research discussion · 6 October 2026

RSA, ML-KEM and an ABW research agenda



The common adoption lesson

- 01 RSA made useful public-key operations practical**
Encryption and signatures gained standard encodings and ecosystem support.
- 02 Kyber balanced competing deployment requirements**
Security confidence, speed and communication cost supported selection.
- 03 ABW needs a complete, quantitatively analysed scheme**
A distinct assumption and inexpensive XORs are a starting point.

RSA: a useful capability package

Public-key encryption

A recipient publishes an encryption key.
The sender encrypts without sharing
the recipient's decryption secret.

Publicly verifiable signatures

A signer uses a private key.
Verifiers use a public key.
No shared signing secret is necessary.

Public keys still need authentication. Encryption and signature roles require suitable schemes.

RSA's compact trapdoor description

$$N = pq \quad ed \equiv 1 \pmod{\lambda(N)}$$

Public operation

$$c = a^e \pmod{N}$$

Trapdoor inverse

$$a = c^d \pmod{N}$$

$\lambda(N) = \text{lcm}(p - 1, q - 1)$. The encoded representative a differs from the application message.

A cheap public operation helps deployment

$$65537 = 2^{16} + 1$$

**16 modular squarings
+ 1 modular multiplication**

This asymmetry favours public encryption and signature verification.

Arithmetic count for the RSA primitive. It is not an end-to-end latency benchmark.

RSA's surrounding standards mattered

Layer	Practical contribution
RSAES-OAEP	Randomized encryption encoding
RSASSA-PSS	A dedicated probabilistic signature scheme
Certificates and key formats	Public-key authentication and interoperability
Libraries and hardware support	Implementations that applications can reuse

The case-study inference: lower integration effort can matter as much as arithmetic elegance.

RSA's advantages have boundaries

Boundary	Consequence
Raw deterministic exponentiation	Textbook RSA lacks semantic encryption security
Private-key computation	Constant-time code, blinding and fault checks matter
Static key transport	Later key compromise can expose recorded sessions
Quantum factoring	RSA is not a post-quantum foundation

TLS 1.3 removed RSA key transport. RSA signatures are a separate role.

CRYSTALS-Kyber and standardized ML-KEM

01 2017: Kyber submission

A concrete CCA-secure KEM candidate with code and test vectors.

02 2022: NIST selection

A multi-round evaluation of security and deployment costs.

03 2024: FIPS 203 publishes ML-KEM

A standardized derivative with its own exact specification.

The Module-LWE design behind Kyber

$$\mathbf{t} = \mathbf{A}\mathbf{s} + \mathbf{e}$$

$$\mathbf{u} = \mathbf{A}^T \mathbf{r} + \mathbf{e}_1$$

$$\mathbf{v} = \mathbf{t}^T \mathbf{r} + \mathbf{e}_2 + \text{Encode}(\mu)$$

$R_q = \mathbb{Z}_{3329}[X]/(X^{256} + 1)$. These equations omit the specified ciphertext compression.

Kyber's engineering choices work together

Design choice	Deployment benefit
Polynomial modules and NTT	Fast arithmetic with compact representations
Public seed for A	Avoid transmitting the entire public matrix
Centered binomial errors	Simple small-error sampling
Quantified compression	Reduce ciphertext size within a decoding budget

Each saving has an accompanying assumption or correctness obligation.

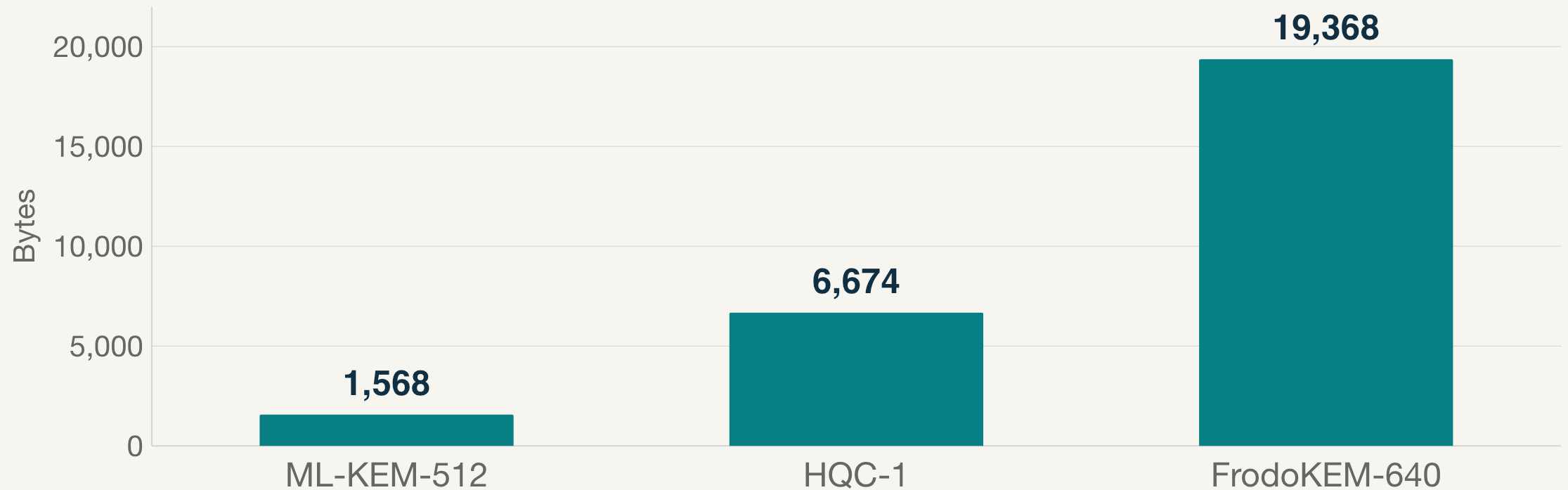
ML-KEM offers concrete deployment sizes

Parameter set	Category	Public key	Ciphertext
ML-KEM-512	1	800 B	768 B
ML-KEM-768	3	1,184 B	1,088 B
ML-KEM-1024	5	1,568 B	1,568 B

Every parameter set outputs a 32-byte shared secret.

FIPS 203, Table 3. Parameter names are not counts of symmetric security bits.

Communication cost is one reason Kyber stands out



Public key + one ciphertext, in bytes. Category 1 examples; security margins and runtime differ.

NIST's choice was a balance of evidence and cost

Comparison	What distinguished the choice
NTRU	Kyber's key generation and broader performance
Saber	Similar efficiency, with more confidence in MLWE evidence
Unstructured LWE	Structure offers substantial efficiency savings
HQC	A code-based alternative serves assumption diversity

NIST assessments in 2022 and 2025. A smaller ciphertext alone does not settle the decision.

A reusable KEM fits existing systems

01 A narrow KeyGen / Encaps / Decaps interface

The KEM supplies a symmetric key for the surrounding protocol.

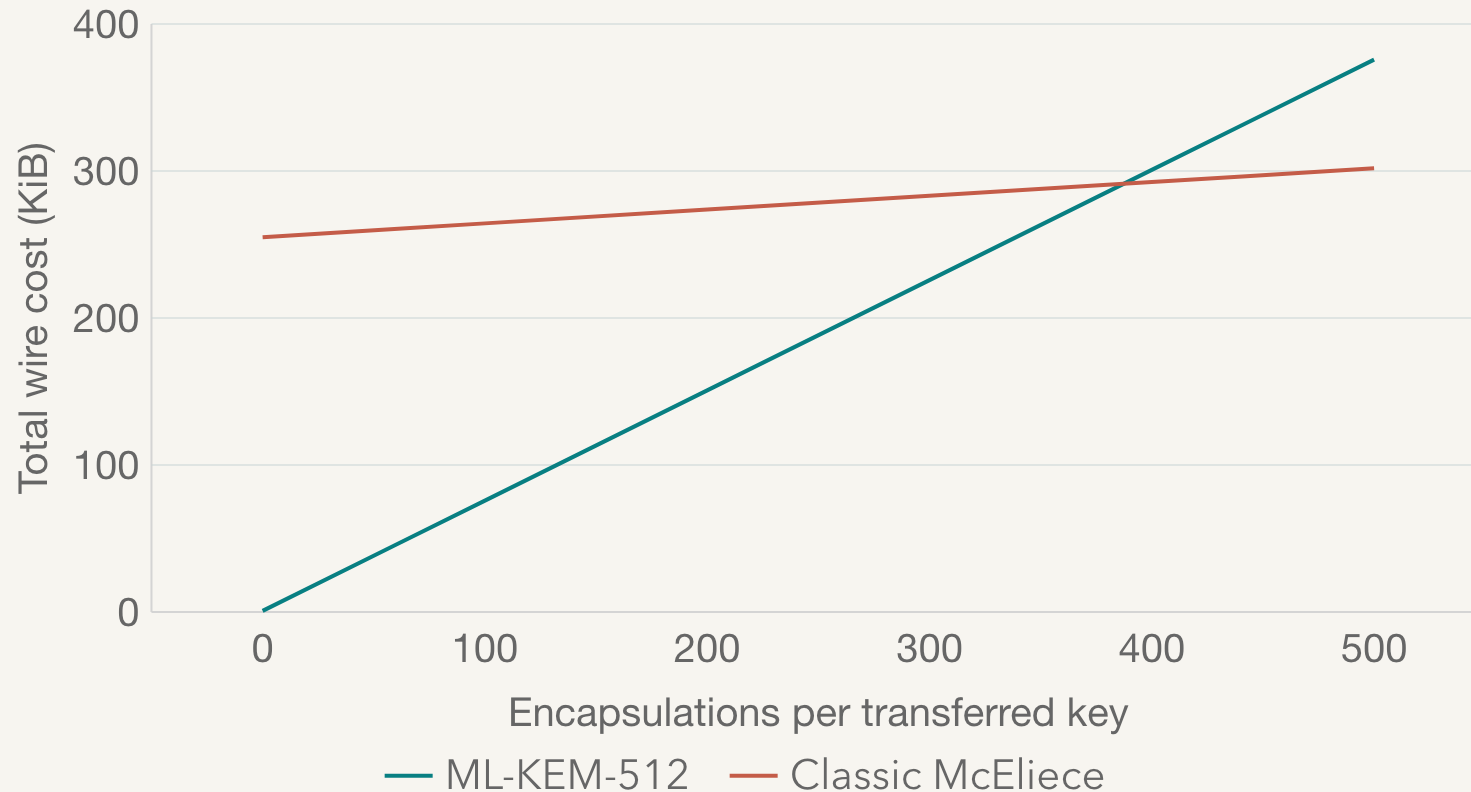
02 Code, test vectors and standard encodings

Applications can reuse implementations, including OpenSSL 3.5 support.

03 Operational and adoption work

Authentication, validation, interoperability and licensing still matter.

The use case can change the communication winner



Cached public keys

Small ciphertexts can
replay
a large initial key transfer.

$$T \geq 388$$

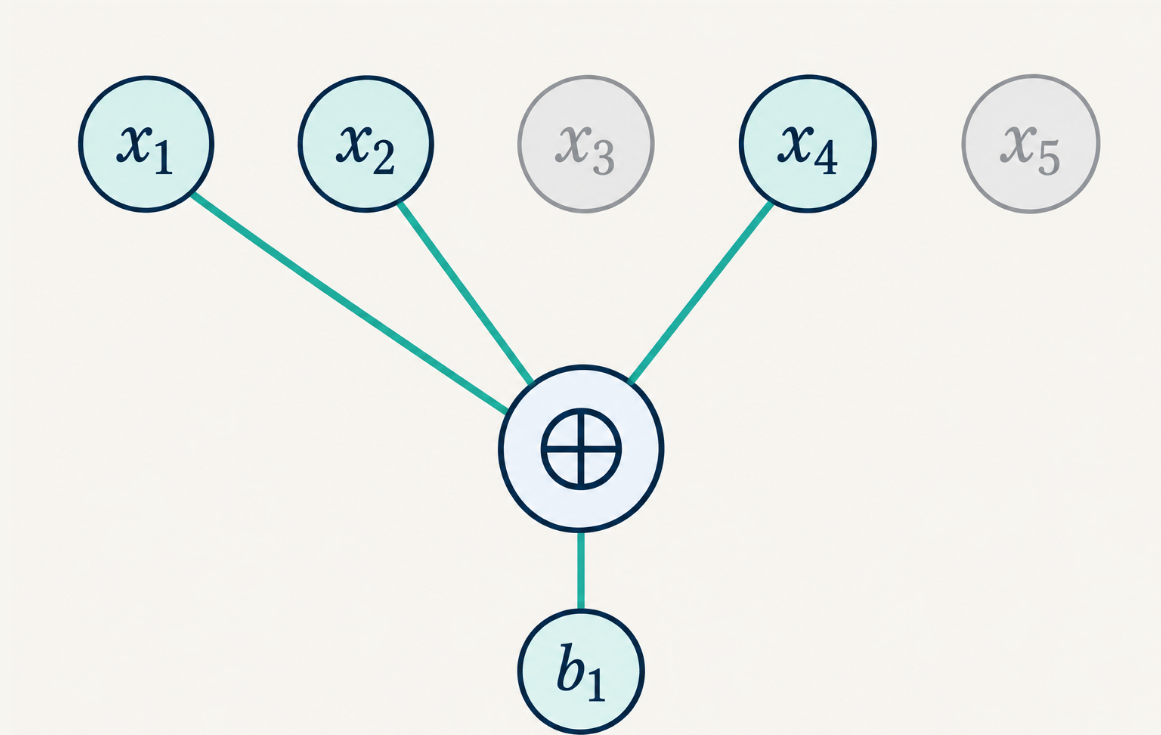
Calculated wire cost only, with equal increments in T. No runtime ranking is implied.

What a new cryptosystem needs to reproduce

Attractive feature	Lesson for an ABW candidate
Clear useful primitive	Start with a complete KEM for one deployment setting
Credible security evidence	Analyse exact samplers and concrete attack costs
Balanced total cost	Count communication, key generation and amplification
Safe integration	Include CCA security, failure handling and implementation

A competitive niche may be a realistic first goal before a general-purpose replacement.

ABW's sparse noisy-equation foundation



$$\mathbf{b} = \mathbf{M}\mathbf{x} + \mathbf{e}$$

Sparse binary rows
Uniform planted $\mathbf{x}$
Independent output flips

Recover $\mathbf{x}$ from $\mathbf{M}$ and $\mathbf{b}$.

Search3LIN: $m = \Theta(n^{1.4})$, $\varepsilon = \Theta(n^{-0.2})$, with suitable constants. Arithmetic is mod 2.

ABW's trapdoor is a short row dependency

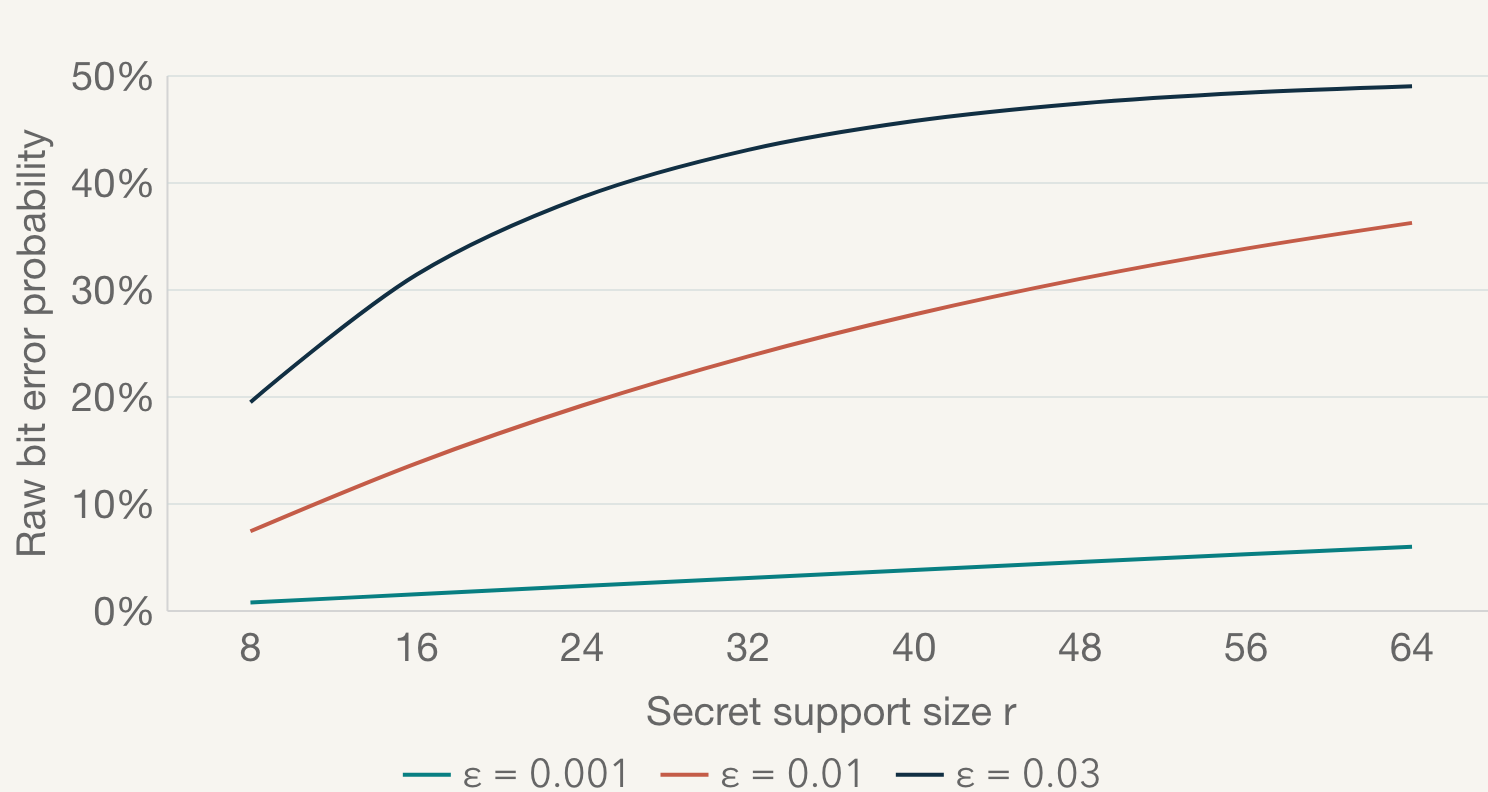
$$\mathbf{s}^T \mathbf{M} = \mathbf{0} \quad \mathbf{s}_m = 1$$

$$\mathbf{c} = \mathbf{M}\mathbf{x} + \mathbf{e} + \mu\mathbf{u}_m$$

$$\mathbf{s}^T \mathbf{c} = \mu + \mathbf{s}^T \mathbf{e}$$

This raw bit mechanism needs correctness and security amplification before becoming a finished primitive.

Correctness is a central cost bottleneck



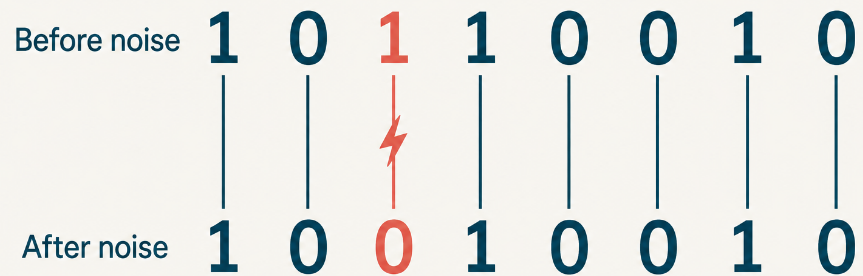
$$P[\text{error}] = \frac{1}{2}[1 - (1 - 2\epsilon)^r]$$

At $r = 32$, $\epsilon = 0.01$

23.8% error

Calculated curves. Complete correctness amplification has real communication and key costs.

Research track 1: cheaper correctness amplification



The possible improvement

Retain useful decoding bias with less repetition or coding overhead.

The security obligation

Keep short relations hidden.
Control correlated error events.

Research proposal. Shorter trapdoors improve correctness and may also help the attacker.

Research track 2: increase payload per ciphertext

$$\mathbf{S}^T \mathbf{M} = \mathbf{0} \quad \mathbf{S}^T \mathbf{G} = \mathbf{I}_\ell$$

$$\mathbf{c} = \mathbf{M}\mathbf{x} + \mathbf{e} + \mathbf{G}\boldsymbol{\mu}$$

$$\mathbf{S}^T \mathbf{c} = \boldsymbol{\mu} + \mathbf{S}^T \mathbf{e}$$

Proposed ℓ -bit interface. A sampler, noise analysis and security proof remain open obligations.

Research track 3: compact public keys

An appealing compression idea

$$M = \text{Expand}(\rho) + \Delta$$

What publishing ρ also reveals

$$\Delta = M - \text{Expand}(\rho)$$

A public base seed can expose planting edits. Compression requires a new distribution analysis.

A competitive scheme needs CCA and quantum analysis

$$\text{Raw Dec}(u_i) = s_i$$

01 A strong CPA foundation

Resolve correctness, repeated ciphertexts and quantitative security.

02 A justified CCA-secure KEM

Analyse rejection, reduction losses and quantum adversaries.

A generic hash of the raw bit output does not supply a secure KEM.

A bounded first project and decision gates

Phase	Concrete result	Decision gate
Weeks 1–3	Exact sampler, assumptions and cost ledger	Can we reproduce the foundation?
Weeks 4–8	Attacks and complete baseline measurements	Is any credible parameter window left?
Weeks 9–12	One focused improvement and proof audit	Does the saving survive the full scheme?

Initial objective: a defensible feasibility result or a specific cryptanalytic finding.

Appendix: RSA arithmetic on a small example

$$p = 61 \quad q = 53 \quad N = 3233$$

$$e = 17 \quad d = 2753$$

$$65^{17} \bmod 3233 = 2790$$

$$2790^{2753} \bmod 3233 = 65$$

Tiny textbook arithmetic only. It provides no practical security.

Appendix: MLWE cancellation and its noise budget

$$\mathbf{v} - \mathbf{s}^T \mathbf{u} = \text{Encode}(\mu) + \mathbf{v}$$

$$\mathbf{v} = \mathbf{e}^T \mathbf{r} + \mathbf{e}_2 - \mathbf{s}^T \mathbf{e}_1$$

The real noise budget also includes compression error.
The full scheme needs a tail bound and failure-safe rejection.

FIPS 203 failure estimates average over honest generation. They do not bound every key separately.

Appendix: the comparison data ledger

Category 1 variant	Public key	Ciphertext	PK + CT
ML-KEM-512	800	768	1,568
HQC-1 (Aug 2025)	2,241	4,433	6,674
FrodoKEM-640 (Sep 2025)	9,616	9,752	19,368
mceliece348864	261,120	96	261,216

All sizes are bytes. Variant and specification dates matter, especially for HQC and FrodoKEM.

Appendix: related directions sharpen the agenda

01 Local PRGs and low-degree analysis

Bogdanov–Kothari–Rosen (2023): a related ABW construction route.

02 Dense-Sparse LPN and lossy trapdoors

Dao–Jain (2024): a changed distribution and different capabilities.

03 High-corruption CSP trapdoors

Hair–Sahai (2026 preprint): a different planting and decoding approach.

References: case-study foundations

Rivest–Shamir–Adleman (1978)

A Method for Obtaining Digital Signatures and Public-Key Cryptosystems

RFC 8017 and RFC 5280

RSA schemes, encoding and the public-key certificate framework

RFC 8446 and Shor (1995 / 1997)

TLS 1.3 and quantum factoring limitations

CRYSTALS-Kyber specification, v3.02 (2021)

Module-LWE design and implementation rationale

NIST FIPS 203 and IR 8413-upd1

ML-KEM parameters and NIST's third-round selection analysis

OpenSSL 3.5 documentation

ML-KEM API and integration support

References: comparisons and the ABW programme

Applebaum–Barak–Wigderson (2009 / STOC 2010)

Public-Key Cryptography from Different Assumptions

Holenstein–Renner (2005) and Hofheinz–Hövelmanns–Kiltz (2017)

Weak encryption amplification and modular FO analysis

NIST IR 8545 and SP 800-227 (2025)

Code-based comparison and KEM use requirements

HQC specification (Aug 2025) and FrodoKEM proposal (Sep 2025)

Exact key and ciphertext sizes for the compared variants

Bogdanov–Kothari–Rosen (2023) and Dao–Jain (2024)

Local PRGs, low-degree evidence and Dense-Sparse LPN

Hair–Sahai (April 2026 preprint)

Public Key Encryption from High-Corruption CSPs